Beta Polarization of Higher Spectral Defects

All-order Stieltjes counterterms, Gamma integrals,
polylogarithmic second jets, and confluent quadratic masses

Research continuation prepared for Vladimir Reshetnikov
ProveIt / Analysis / Polylogarithms
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Abstract

We resolve the explicit higher-defect and two-mass second-derivative questions posed in the quadratic-spectrum continuation of the ProveIt programme. An endpoint-subtracted Euler beta integral gives every spectral defect of order r as a finite combination of ordinary logarithmically weighted integrals of linear Hurwitz-zeta derivatives of orders at most $r-1$. A separate, normally convergent coefficient formula makes all resonant generalized Stieltjes terms explicit, with coefficients in generalized harmonic numbers.

At second order the nonlocal part is a Gamma–Hurwitz chord integral and the correction is a finite odd-weight polynomial. In contrast to the first-derivative parity theorem, no nonzero polynomial weight annihilates the second defect for every mass. The scalar case gives convergent arctangent-square, log-Gamma, harmonic-zeta, and polygamma identities. For every nonpositive integral spectral argument, the bilateral second jet is expressed by finitely many complex-shift Hurwitz second derivatives and one ordinary integral of an explicit finite polylogarithm expression.

For any number of quadratic factors with an even polynomial weight, the second jet has a pairwise covariance formula. Each covariance is a squared-mass beta chord of the already evaluated first jet; thus it has finite Gamma or polylogarithmic integrands. Its full confluent expansion has coefficients $2(-1)^k H_{k-1} Z_W(k)/k$, with finite polygamma reductions. All parameter primitives are anchored explicitly. The article includes proofs, executable exact checks, numerical diagnostics, an integration fragment, and a scoped provenance audit. The beta identity and first-derivative anomaly mechanism are classical inputs, not priority claims; minimal-function-algebra questions remain separate.

Status. Ordinary analytic proofs with reproducible finite checks; not proof-assistant formalized or independently peer reviewed. No repository files were modified.

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1 The research question and the contribution

The canonical ProveIt manuscript distinguishes exact analytic identities, relations in a specified formal presentation, and numerical residuals [1]. This distinction is particularly important for zeta-regularized logarithms: a pointwise logarithmic identity need not survive spectral continuation without a correction.

The immediate predecessor, *Quadratic-Spectrum Stieltjes Identities* [2], proves a first-derivative factorization formula and asks in its Section 7.6 for three further developments. Question 2 asks for an explicit separation of the nonlocal and residue parts of higher factorization defects. Question 7 asks for the second derivative of a product of two quadratic spectra, including its coincident-mass limit, without an unspecified subtraction of divergent sums. Question 1 asks for useful function coordinates for higher derivatives at negative integers, and separately whether a smaller finite function algebra suffices.

Theorems 2.3 and 3.1 give all-order formulas for the first problem. Theorems 4.1 and 4.2 give a particularly explicit second-order answer and its exact universal vanishing classification. Theorems 7.2 and 7.3 solve the two-mass representation and confluence problem, and also handle arbitrarily many masses. Theorem 6.1 supplies a finite-integrand representation at every negative integer. It does *not* prove that the remaining ordinary integral is irreducible to a smaller algebra.

The main mechanism is order lowering. The subtracted beta kernel has a simple zero in the spectral parameter at zero. Consequently an rth defect needs linear Hurwitz derivatives only through order $r-1$ inside the integral. At order two these are first Hurwitz derivatives, including $\log\Gamma$. An independent inverse-power expansion identifies exactly where generalized Stieltjes constants replace ordinary zeta values.

1.1 Classical inputs and the priority boundary

Euler's beta integral, its two-denominator parameterization, Gamma products, Hurwitz continuation and differentiation, and elementary polylogarithm relations are classical [4, 5, 6, 7]. The multiplicative anomaly for first derivatives of spectral zeta functions is also an established subject; Dowker explicitly treats products of linear and second-order factors [8]. The predecessor's finite first-jet polylogarithm formulas are retained with attribution, not relabeled as new results.

The contribution here is the precise endpoint-subtracted all-jet calculus, its harmonic/Stieltjes coefficient implementation, the second universal-vanishing theorem, and the explicit confluent and primitive consequences. No comprehensive historical-priority or arithmetic independence claim is made. Neither the Gaussian S_6 nor the revised S_8 special-value problem is solved by this article.

1.2 Source identity and scope of inspection

The predecessor's single-file TeX was read in full for the relevant sections and materialized for comparison. Its SHA-256 is

`4185cc4aa2299c8f417ca376b3ddc4af3fab58752d4cd93f70c2afad661f95aa`.

This agrees exactly with the repository's sixteenth-intake manifest at `5a5c0ed6573d25fcabf93653d72449f76df2df79`. The recorded path is

`Analysis/Polylogarithms/docs/reports/stieltjes-correlation-calculus/continuations/quadratic-spectrum-stieltjes/article_singlefile.tex`.

Thus the mathematical target is a byte-identified repository report, not a conjecture inferred from a filename. The canonical preface, differentiation and research-status sections, incoming catalogue, and relevant intake metadata were also inspected. This was a targeted audit, not an exhaustive reading of all incoming reports or the entire repository. Detailed source records are included in `notes/SOURCE_AUDIT.md`.

2 Analytic normalization and an endpoint-subtracted beta master

Fix $a > 0$, write $x_n = n + a$, and let

$$W(x) = \sum_{d=0}^{D} w_d x^d \in \mathbb{C}[x]. \tag{2.1}$$

Coefficients outside this range are zero. All spectral superscripts (r) below differentiate the spectral variable, not the shift. On the right half-plane, Log and $\log\Gamma$ mean their holomorphic branches that are real on the positive axis. In particular, $\log\Gamma(z)$ is not formed by taking the principal logarithm of a numerically evaluated $\Gamma(z)$.

For $\operatorname{Re}(a+z) > 0$, define initially for $\operatorname{Re} s > D+1$

$$F_{a,W}(s;z) = \sum_{n\geq 0} W(x_n)(x_n+z)^{-s}. \tag{2.2}$$

A finite binomial expansion gives its continuation:

$$F_{a,W}(s;z) = \sum_{d=0}^{D} w_d \sum_{h=0}^{d} \binom{d}{h} (-z)^{d-h} \zeta(s-h, a+z). \tag{2.3}$$

It is holomorphic at $s=0$, locally uniformly in z in the stated domain. Put

$$\Phi_{a,W}(z) = F'_{a,W}(0;z), \qquad A_W(z) = \sum_{d=0}^{D} \frac{(-1)^{d+1} w_d z^{d+1}}{d+1}. \tag{2.4}$$

For $W = 1$ the classical Lerch formula gives $\Phi_{a,1}(z) = \log\Gamma(a+z) - \frac{1}{2}\log(2\pi)$.

For endpoints u, v with $\operatorname{Re}(a+u), \operatorname{Re}(a+v) > 0$, define

$$Z^{u,v}_{a,W}(s) = \sum_{n\geq 0} W(x_n) \exp\{-s(\operatorname{Log}(x_n+u) + \operatorname{Log}(x_n+v))\}. \tag{2.5}$$

The definition uses the *sum* of the two specified logarithms. Its initial convergence domain is $2\operatorname{Re} s > D+1$. Set

$$G_{a,W}(s;u,v) = \tfrac{1}{2}\{F_{a,W}(2s;u) + F_{a,W}(2s;v)\} - Z^{u,v}_{a,W}(s), \tag{2.6}$$

$$\Delta_r(a,W;u,v) = G^{(r)}_{a,W}(0;u,v). \tag{2.7}$$

For $u = ib, v = -ib$, write $Z_W(s;a,b)$ and $\Delta_r(W;a,b)$. Then $Z_W(s;a,b) = \sum W(x_n)(x_n^2+b^2)^{-s}$, and

$$\Delta_r(W;a,b) = 2^{r-1}(F^{(r)}_{a,W}(0;ib) + F^{(r)}_{a,W}(0;-ib)) - Z^{(r)}_W(0;a,b). \tag{2.8}$$

For an analytic function f on a neighborhood of the segment $[u, v]$, define the symmetric endpoint subtraction

$$\mathcal{C}_t f(u, v) = f(u) + f(v) - f((1-t)u + tv) - f(tu + (1-t)v). \tag{2.9}$$

This vanishes to first order at each endpoint of $0 \le t \le 1$. More precisely,

$$\lim_{t \to 0,1} \frac{\mathcal{C}_t f(u, v)}{t(1-t)} = (v-u)(f'(v) - f'(u)). \tag{2.10}$$

The limit also holds at $u = v$, where both sides are zero.

Lemma 2.1 (The zero jet is a local polynomial)**.** *For all admissible a, z, W,*

$$F_{a,W}(0; z) = \sum_{d=0}^{D} w_d \zeta(-d, a) + A_W(z). \tag{2.11}$$

In particular, $\mathcal{C}_t F_{a,W}(0; \cdot) = \mathcal{C}_t A_W$ is independent of a.

Proof. For $|z| < a$, expand the summand binomially and continue:

$$F_{a,W}(s; z) = \sum_{d=0}^{D} w_d \sum_{k \ge 0} \frac{(-1)^k (s)_k z^k}{k!} \zeta(s + k - d, a). \tag{2.12}$$

The $k = 0$ term gives $\zeta(-d, a)$. Every $k \ge 1$ term has a factor s and vanishes at zero except $k = d + 1$, where the simple Hurwitz pole cancels it. Since $(s)_k / k! = s/k + O(s^2)$, that surviving term is $(-1)^k z^k / k$. The series and its pole-subtracted terms converge locally normally for $|z| < a$; the ratio $|z|/a < 1$ dominates the polynomial coefficient growth. This proves the formula locally. Both sides are holomorphic on $\mathrm{Re}(a + z) > 0$, so analytic continuation proves it there. □

Theorem 2.2 (Subtracted beta master)**.** *The function $Z^{u,v}_{a,W}$ has a continuation holomorphic at zero. As an identity of germs at $s = 0$,*

$$G_{a,W}(s; u, v) = \frac{s}{4} B_*(s) \int_0^1 \frac{[t(1-t)]^s}{t(1-t)} \mathcal{C}_t F_{a,W}(2s; \cdot)(u, v) \, \mathrm{d}t, \tag{2.13}$$

where

$$B_*(s) = \frac{\Gamma(1 + 2s)}{\Gamma(1 + s)^2}. \tag{2.14}$$

The integral in (2.13) *is an ordinary convergent integral near zero, jointly holomorphic in s, u, v in their stated local domains. Every fixed spectral derivative may be passed under it.*

Proof. For $\mathrm{Re}\, s > (D + 1)/2$ use the classical two-denominator beta identity, which follows from Euler's beta integral or two Gamma integrals:

$$A^{-s} B^{-s} = \frac{\Gamma(2s)}{\Gamma(s)^2} \int_0^1 \frac{t^{s-1}(1-t)^{s-1}}{((1-t)A + tB)^{2s}} \, \mathrm{d}t.$$

It holds for $\mathrm{Re}\, A, \mathrm{Re}\, B > 0$ with the prescribed logarithms. The absolute convergence of the n-sum is uniform in t, apart from the integrable beta weight. Sum it with $A = x_n + u$, $B = x_n + v$, and

average the result under $t \mapsto 1-t$. Since the beta weight integrates to $\Gamma(s)^2/\Gamma(2s)$, subtracting the endpoint average produces

$$G(s) = \frac{\Gamma(2s)}{2\Gamma(s)^2} \int_0^1 t^{s-1}(1-t)^{s-1} \mathcal{C}_t F(2s;\cdot)\, \mathrm{d}t.$$

The prefactor equals $sB_*(s)/4$.

For a small closed disk about $s = 0$, the finite formula (2.3) is holomorphic, uniformly on the compact endpoint segments under consideration. Taylor expansion in t gives $\mathcal{C}_t F(2s;\cdot) = O(t)$ at zero and $O(1-t)$ at one, uniformly on that disk. Each fixed s-derivative adds only a fixed power of $|\log(t(1-t))|$. Thus the integral and all such derivatives are absolutely convergent and locally dominated whenever $\operatorname{Re} s > -1$ in this small disk. This proves holomorphy and differentiation under the integral.

The right side continues meromorphically from the initial half-plane, with the isolated positive spectral poles of the finite Hurwitz expression bypassed. It is holomorphic at zero. Defining Z by (2.6) there proves both its continuation and the identity. The argument is locally uniform in u, v and covers coincident endpoints. □

2.1 All defects as finite lower-jet integrals

Write

$$B_*(s) = \sum_{m\geq 0} \beta_m s^m = \exp\left(\sum_{n\geq 2} \frac{(-1)^n (2^n - 2)\zeta(n)}{n} s^n \right). \tag{2.15}$$

Hence $\beta_0 = 1$, $\beta_1 = 0$, $\beta_2 = \zeta(2)$, $\beta_3 = -2\zeta(3)$, and $\beta_4 = \frac{7}{2}\zeta(4) + \frac{1}{2}\zeta(2)^2$. An exact finite recurrence is

$$m\beta_m = \sum_{n=2}^{m} (-1)^n (2^n - 2)\zeta(n)\beta_{m-n}. \tag{2.16}$$

Put $\lambda(t) = \log(t(1-t))$ and

$$I_{j,p}(a, W; u, v) = \int_0^1 \frac{\lambda(t)^p}{t(1-t)} \mathcal{C}_t(F^{(j)}_{a,W}(0;\cdot))(u, v)\, \mathrm{d}t. \tag{2.17}$$

No finite-part convention is involved in this integral.

Theorem 2.3 (All-order lower-jet reduction)**.** *For every integer $r \geq 1$ and every admissible endpoint pair,*

$$\boxed{\Delta_r(a, W; u, v) = \frac{r!}{4} \sum_{m+j+p=r-1} \frac{2^j \beta_m}{j!p!} I_{j,p}(a, W; u, v).} \tag{2.18}$$

All sums are finite. The maximal spectral derivative inside an integral is $r-1$. Every $j = 0$ term is an explicit polynomial in u, v independent of a.

Proof. Expand $B_*(s)$, $\exp(s\lambda(t))$ and $F(2s; z) = \sum_{j\geq 0} 2^j s^j F^{(j)}(0; z)/j!$ in (2.13). The domination established in Theorem 2.2 justifies extraction of each coefficient under the integral. The outer factor $s/4$ gives exactly the constraint $m + j + p = r - 1$. Lemma 2.1 identifies the local terms. □

The first four rows, suppressing common arguments, are

$$\Delta_1 = \tfrac{1}{4} I_{0,0}, \tag{2.19}$$
$$\Delta_2 = I_{1,0} + \tfrac{1}{2} I_{0,1}, \tag{2.20}$$
$$\Delta_3 = 3I_{2,0} + 3I_{1,1} + \tfrac{3}{4} I_{0,2} + \tfrac{3}{2}\zeta(2) I_{0,0}, \tag{2.21}$$
$$\Delta_4 = 8I_{3,0} + 12I_{2,1} + 6I_{1,2} + I_{0,3} + 12\zeta(2) I_{1,0} + 6\zeta(2) I_{0,1} - 12\zeta(3) I_{0,0}. \tag{2.22}$$

At conjugate endpoints and for even W, the polynomial A_W is odd, so $\mathcal{C}_t A_W(ib, -ib) = 0$ identically. All $I_{0,p}$ then vanish. This removes the local polynomial terms but does not remove the positive-j integrals. That distinction is the reason the first-derivative parity theorem does not extend unchanged to higher jets.

3 A global harmonic–Stieltjes coefficient formula

The beta master is finite in the number of function coordinates at every order. A different representation is useful for exact coefficients and for evaluating resonances without subtracting two large meromorphic terms. Use the convention

$$\zeta(1+\varepsilon, X) = \frac{1}{\varepsilon} + \sum_{n \geq 0} \frac{(-1)^n \gamma_n(X)}{n!} \varepsilon^n, \qquad \gamma_0(X) = -\psi(X). \tag{3.1}$$

Let

$$O_k^{(p)} = \sum_{j=0}^{k-1} \frac{1}{(2j+1)^p} = H_{2k}^{(p)} - 2^{-p} H_k^{(p)}, \qquad O_k = O_k^{(1)}. \tag{3.2}$$

The coefficient polynomial controlling the conjugate defect is

$$D_k(s) = \frac{(2s)_{2k}}{(2k)!} - \frac{(s)_k}{k!} = \sum_{\ell \geq 2} d_{k,\ell} s^\ell \tag{3.3}$$
$$= \frac{s}{k} \prod_{j=1}^{k-1} \left(1 + \frac{s}{j}\right) \left\{ \prod_{j=0}^{k-1} \left(1 + \frac{2s}{2j+1}\right) - 1 \right\}. \tag{3.4}$$

The second equality follows from duplication of the rising factorial. It exhibits the double zero without cancellation in floating-point arithmetic. In particular,

$$d_{k,2} = \frac{2O_k}{k}, \tag{3.5}$$
$$d_{k,3} = \frac{2}{k}(O_k H_{k-1} + O_k^2 - O_k^{(2)}). \tag{3.6}$$

Every higher coefficient is a finite polynomial in generalized harmonic numbers. One can either multiply the finite products in (3.4), or use

$$\log \prod_{j=1}^{k-1} (1 + s/j) = \sum_{p \geq 1} \frac{(-1)^{p-1}}{p} H_{k-1}^{(p)} s^p, \qquad \log \prod_{j=0}^{k-1} (1 + 2s/(2j+1)) = \sum_{p \geq 1} \frac{(-1)^{p-1} 2^p}{p} O_k^{(p)} s^p.$$

Only finitely many coefficients are needed for a prescribed jet.

For $r \geq 1$ define the elementary finite-head term

$$q_r(x,b) = 2^{r-1}\{(-\operatorname{Log}(x+ib))^r + (-\operatorname{Log}(x-ib))^r\} - (-\log(x^2+b^2))^r. \tag{3.7}$$

Thus $q_1 = 0$ and

$$q_2(x,b) = \big(\operatorname{Log}(x+ib) - \operatorname{Log}(x-ib)\big)^2 = -4\arctan^2(b/x). \tag{3.8}$$

Theorem 3.1 (Every resonant coefficient, with a global finite head)**.** *Let $a, b > 0$, $r \geq 1$, and choose any integer $N \geq 0$ such that $X = a + N > b$. For an integer τ put*

$$T_{r,k}(\tau; X) = r! \sum_{\ell=2}^{r} \frac{2^{r-\ell} d_{k,\ell}}{(r-\ell)!} \zeta^{(r-\ell)}(\tau, X), \qquad \tau \neq 1, \tag{3.9}$$

and at the resonance set

$$T_{r,k}(1; X) = r! \left\{ \frac{d_{k,r+1}}{2} + \sum_{\ell=2}^{r} \frac{(-2)^{r-\ell} d_{k,\ell}}{(r-\ell)!} \gamma_{r-\ell}(X) \right\}. \tag{3.10}$$

Empty sums are zero. Then

$$\boxed{\Delta_r(W; a, b) = \sum_{n=0}^{N-1} W(a+n) q_r(a+n, b) + \sum_{d=0}^{D} w_d \sum_{k \geq 1} (-1)^k b^{2k} T_{r,k}(2k-d; X).} \tag{3.11}$$

The double sum is absolutely and locally normally convergent. It is independent of the permissible cutoff N. The largest Stieltjes index which can occur is $r-2$.

Proof. On the tail $x \geq X > b$, expand $\frac{1}{2}\{(x+ib)^{-2s} + (x-ib)^{-2s}\}$ and $(x^2+b^2)^{-s}$ in powers of b/x. Odd powers cancel in the first expression. Subtracting gives

$$\begin{aligned} G(s; ib, -ib) = &\sum_{n<N} \tfrac{1}{2} W(x_n)\{(x_n+ib)^{-2s} + (x_n-ib)^{-2s} - 2(x_n^2+b^2)^{-s}\} \\ &+ \sum_{d=0}^{D} \sum_{k \geq 1} w_d (-1)^k b^{2k} D_k(s) \zeta(2s+2k-d, X), \end{aligned} \tag{3.12}$$

where $0 \leq d \leq D$ and $k \geq 1$. For k sufficiently large, the Hurwitz argument has large positive real part. Uniformly for s on a fixed small circle, $\zeta(2s+2k-d, X)$ is bounded by a constant times $X^{-2k+d-2\operatorname{Re} s}$, up to a harmless polynomial factor. The rising factorial ratios grow only polynomially in k on such a circle. Therefore $(b/X)^{2k}$ dominates the tail normally. A fixed number of spectral or shift derivatives introduces at most polynomial and logarithmic factors. The finitely many possible Hurwitz poles are handled individually.

If $2k - d \neq 1$, Taylor multiplication gives (3.9). At $2k - d = 1$, insert (3.1) with $\varepsilon = 2s$. Because D_k starts at s^2, the product is holomorphic; its rth derivative is exactly (3.10). The finite head differentiates to Wq_r. This proves the formula and all convergence assertions. Different cutoffs give the same continued function before differentiation, proving cutoff independence. □

For clarity, the first three resonant rows are

$$T_{1,k}(1; X) = \tfrac{1}{2} d_{k,2}, \tag{3.13}$$

$$T_{2,k}(1; X) = d_{k,3} + 2d_{k,2}\gamma_0(X), \tag{3.14}$$

$$T_{3,k}(1; X) = 3d_{k,4} + 6d_{k,3}\gamma_0(X) - 12d_{k,2}\gamma_1(X). \tag{3.15}$$

The term $d_{k,r+1}/2$ is easily lost by truncating D_k at order r before multiplying by the pole. Such a truncation gives an incorrect jet even though $D_k(0) = D_k'(0) = 0$.

At $r = 1$, only $d = 2k - 1$ contributes, and (3.11) recovers the predecessor's identity

$$\Delta_1(W; a, b) = \sum_{k\geq 1} \frac{(-1)^k w_{2k-1} b^{2k}}{k} O_k. \tag{3.16}$$

This is inherited, not a new first-order theorem. At higher orders, the same finite resonance set remains, but infinitely many nonresonant coordinates survive. The resonant Stieltjes values in (3.10) should not be confused with the shift-independent polynomial part of the beta-integral formula: the two formulas organize the same continued expression in different bases.

4 The exact second defect and its vanishing classification

Define the Gamma–Hurwitz polarization

$$\mathcal{K}_{a,W}(u, v) = \int_0^1 \frac{\mathcal{C}_t \Phi_{a,W}(u, v)}{t(1-t)} \, \mathrm{d}t. \tag{4.1}$$

For a fixed polynomial weight it is one ordinary integral of the finite Hurwitz expression (2.3). For $W = 1$ it involves only $\log\Gamma$. Its integrand has removable endpoints by (2.10); it is jointly holomorphic and vanishes when $u = v$.

Theorem 4.1 (Gamma integral plus an explicit odd-weight polynomial)**.** *For every polynomial W and every $a, b > 0$,*

$$\boxed{\Delta_2(W; a, b) = \mathcal{K}_{a,W}(ib, -ib) - C_W(b), \qquad C_W(b) = \sum_{k\geq 1} \frac{2(-1)^k w_{2k-1} b^{2k}}{k} (O_k^2 + O_k^{(2)}).} \tag{4.2}$$

The correction is finite and independent of a. In particular,

$$Z_W''(0; a, b) = 2\{F_{a,W}''(0; ib) + F_{a,W}''(0; -ib)\} - \mathcal{K}_{a,W}(ib, -ib) + C_W(b). \tag{4.3}$$

There is no undefined regularized sum on the right side.

Proof. The second beta row gives $\Delta_2 = \mathcal{K} + \frac{1}{2} I_{0,1}$. For conjugate endpoints, Lemma 2.1 gives

$$\mathcal{C}_t A_W(ib, -ib) = \sum_{k\geq 1} \frac{(-1)^k w_{2k-1} b^{2k}}{k} (1 - (1-2t)^{2k}).$$

We evaluate the logarithmic moment, including its normalization. Put

$$P_k(s) = \prod_{j=0}^{k-1} \left(1 + \frac{2s}{2j+1}\right), \qquad J_k(s) = \int_0^1 [t(1-t)]^{s-1} (1 - (1-2t)^{2k}) \, \mathrm{d}t.$$

By $y = 1 - 2t$ and Euler's beta integral,

$$J_k(s) = \frac{2}{sB_*(s)} (1 - P_k(s)^{-1}). \tag{4.4}$$

This is first obtained for $\operatorname{Re} s > 0$ and is holomorphic at zero; its defining subtracted integral is locally integrable for $\operatorname{Re} s > -1$. Since

$$P_k(s)^{-1} = 1 - 2O_k s + 2(O_k^2 + O_k^{(2)})s^2 + O(s^3), \qquad B_*(s) = 1 + O(s^2),$$

we have $J_k(0) = 4O_k$ and $J_k'(0) = -4(O_k^2 + O_k^{(2)})$. Thus $\frac{1}{2}I_{0,1} = -C_W(b)$. The second formula follows directly from the definition of Δ_2. □

The first correction terms in $\Delta_2 - \mathcal{K}$ are

$$4w_1b^2 - \frac{26}{9}w_3b^4 + \frac{1576}{675}w_5b^6 - \frac{21946}{11025}w_7b^8 + \cdots . \tag{4.5}$$

For real weights, an equivalent real integral is

$$\mathcal{K}_{a,W}(ib, -ib) = 8\int_0^1 \frac{\operatorname{Re}\Phi_{a,W}(ib) - \operatorname{Re}\Phi_{a,W}(ibt)}{1-t^2}\,\mathrm{d}t. \tag{4.6}$$

For complex weights, use (4.1) rather than pulling coefficients through a real-part operation.

4.1 An independent coefficient derivation

A useful consistency check on the beta proof is the Taylor expansion

$$\Phi_{a,W}(z) = \Phi_{a,W}(0) + \sum_{j\ge1}\frac{(-1)^j z^j}{j}\left\{\sum_{d=0}^{D} w_d\widehat{\zeta}(j-d,a) + w_{j-1}H_{j-1}\right\}, \quad |z| < a, \tag{4.7}$$

where $\widehat{\zeta}(1,a) = \gamma_0(a)$ and $\widehat{\zeta}(s,a) = \zeta(s,a)$ at the other displayed integer arguments. To prove it, differentiate (2.12); the resonant term uses $(s)_j/j! = s(1 + H_{j-1}s + O(s^2))/j$. The harmonic addend is essential.

For conjugate endpoints, direct polynomial integration gives

$$\int_0^1 \frac{\mathcal{C}_t(z^{2k})(ib,-ib)}{t(1-t)}\,\mathrm{d}t = 8(-1)^k O_k b^{2k}, \qquad \mathcal{C}_t(z^{2k+1})(ib,-ib) = 0.$$

Inserting (4.7) into $\mathcal{K}$ and using $H_{2k-1} = O_k + \frac{1}{2}H_{k-1}$ reproduces exactly (4.2) from the second Stieltjes row. This explains why an uncorrected termwise "sum of logarithm squares" can differ from $\mathcal{K}$ for an odd polynomial weight.

Theorem 4.2 (Universal vanishing at second order)**.** *Fix $a > 0$ and $W \in \mathbb{C}[x]$. Then*

$$\Delta_2(W; a, b) = 0 \text{ for every } b > 0 \quad \Longleftrightarrow \quad W = 0. \tag{4.8}$$

In fact, for a nonzero W, all sufficiently high even Taylor coefficients in b at zero are nonzero.

Proof. For $0 < b < a$ use (3.11) with $N = 0$. When $2k > D + 1$, there is no resonance, and the coefficient of b^{2k} is

$$\frac{4(-1)^k O_k}{k}\sum_{d=0}^{D} w_d\zeta(2k-d,a) = \frac{4(-1)^k O_k}{k}\sum_{n\ge0}\frac{W(a+n)}{(a+n)^{2k}}. \tag{4.9}$$

Suppose $W \neq 0$, and let n_0 be the smallest nonnegative integer with $W(a+n_0) \neq 0$. It exists because a nonzero polynomial has finitely many roots. Put $x_0 = a + n_0$. Multiplying the last sum by x_0^{2k} gives

$$W(x_0) + \sum_{n>n_0} W(a+n) \left(\frac{x_0}{a+n}\right)^{2k} \longrightarrow W(x_0) \neq 0.$$

To justify the limit for complex coefficients as well, fix k_0 with $2k_0 > D+1$. The absolute values for $k \geq k_0$ are dominated by a convergent polynomially weighted Dirichlet series. Dominated convergence then applies. Thus the coefficients in (4.9) are nonzero for every sufficiently large k, contradicting an identically zero analytic function. The converse is immediate. □

This theorem concerns an identity valid for all masses, not zeros at one chosen mass. It also does not assert positivity for arbitrary weights: continued polynomially weighted sums can have either sign. The exact contrast is that even W annihilates *every first defect*, whereas only $W = 0$ annihilates *every second defect*.

5 Ordinary Gamma, polygamma, and harmonic identities

The scalar specialization needs no regularization at all. It provides both concrete identities and a direct way to see the nonlocal term.

Proposition 5.1 (A convergent logarithm-square polarization)**.** *For $a > 0$ and $\mathrm{Re}(a+u), \mathrm{Re}(a+v) > 0$,*

$$\mathcal{K}_{a,1}(u,v) = \sum_{n\geq 0} \big(\mathrm{Log}(n+a+u) - \mathrm{Log}(n+a+v)\big)^2. \tag{5.1}$$

The series is absolutely and locally normally convergent. The square is an ordinary complex square, not an absolute square. Its mixed derivative is

$$\partial_u \partial_v \mathcal{K}_{a,1}(u,v) = -2\frac{\psi(a+v) - \psi(a+u)}{v-u}, \tag{5.2}$$

with diagonal value $-2\psi'(a+u)$.

Proof. For fixed nonzero A, B in the right half-plane, the scalar beta formula of Theorem 2.2, without any sum, gives on taking second jets

$$\int_0^1 \frac{\mathcal{C}_t(-\mathrm{Log}(x+z))(u,v)}{t(1-t)}\,\mathrm{d}t = \big(\mathrm{Log}(x+u) - \mathrm{Log}(x+v)\big)^2.$$

Equivalently this follows by two elementary differentiations in u, v and evaluation on the diagonal. In the Weierstrass expansion of $\log\Gamma(a+z)$, the affine terms in z are annihilated by $\mathcal{C}_t$. The remaining tail is $O((n+a)^{-2})$, uniformly on compact endpoint sets, including after the displayed endpoint subtraction. Thus the scalar identity may be summed under the integral, proving (5.1). Differentiating the normally convergent series and using the convergent digamma difference proves (5.2). The limit at the diagonal follows by continuity. □

The collision expansion of this Gamma identity is particularly simple. For admissible endpoints and $|\delta| < |a+u|$,

$$\mathcal{K}_{a,1}(u, u+\delta) = \sum_{k\geq 2} \frac{2(-1)^k H_{k-1}}{k} \delta^k \zeta(k, a+u) \tag{5.3}$$

$$= 2\sum_{k\geq 2} \frac{H_{k-1}}{k!} \delta^k \psi^{(k-1)}(a+u). \tag{5.4}$$

Indeed, $\log^2(1+z) = \sum_{k\geq 2} 2(-1)^k H_{k-1} z^k/k$ for $|z| < 1$. The endpoint hypothesis fixes the logarithm; since $\mathrm{Re}(a+u) > 0$, $|n+a+u| \geq |a+u|$, which justifies normal summation of this series. The first terms are

$$\delta^2 \zeta(2, a+u) - \delta^3 \zeta(3, a+u) + \frac{11}{12}\delta^4 \zeta(4, a+u) + O(\delta^5).$$

For $a, b > 0$, put $S_a(b) = \sum_{n\geq 0} \arctan^2(b/(n+a))$.

Corollary 5.2 (Arctangent–Gamma identities)**.** *The following identities hold:*

$$\Delta_2(1; a, b) = -4S_a(b), \tag{5.5}$$

$$S_a(b) = 2b \int_0^b \frac{\mathrm{Re}\log\Gamma(a+iy) - \mathrm{Re}\log\Gamma(a+ib)}{b^2 - y^2}\,\mathrm{d}y, \tag{5.6}$$

$$S_a(b) = \sum_{k\geq 1} \frac{(-1)^{k-1} O_k}{k} b^{2k} \zeta(2k, a), \qquad 0 < b < a. \tag{5.7}$$

For arbitrary b, the global version of the last formula is obtained by adding $\sum_{n<N} \arctan^2(b/(a+n))$ and replacing a by $a+N > b$ in its zeta tail.

Proof. Apply (5.1) to $u = ib, v = -ib$, use (3.8), and then use (4.6) with $y = bt$. The last identity follows either by the Taylor series of $\arctan^2$ or by (3.11) for $W = 1, r = 2$. □

In particular the constant weight, which has zero first defect, has a strictly negative second defect at every positive mass. This is an exact counterexample to an unqualified extension of the first-order parity law. The predecessor already warned against that extension; this corollary supplies the explicit obstruction rather than correcting a false claim in that report.

At an integral shift $a = N+1$, the same identities become identities in ordinary harmonic tails, since

$$\zeta(2k, N+1) = \zeta(2k) - H_N^{(2k)}.$$

For example, on $0 < b < N+1$,

$$2b \int_0^b \frac{\mathrm{Re}\log\Gamma(N+1+iy) - \mathrm{Re}\log\Gamma(N+1+ib)}{b^2 - y^2}\,\mathrm{d}y$$
$$= \sum_{k\geq 1} \frac{(-1)^{k-1}}{k} \left(H_{2k} - \frac{1}{2}H_k\right) b^{2k} (\zeta(2k) - H_N^{(2k)}). \tag{5.8}$$

This is an equality of ordinary convergent quantities with every coefficient explicit.

5.1 Reflection produces polylogarithms of order one

For $0 < a < 1$, Gamma reflection gives

$$\operatorname{Re}\log\Gamma(a+iy) + \operatorname{Re}\log\Gamma(1-a+iy) = \log(2\pi) - \tfrac{1}{2}\log\bigl(2(\cosh(2\pi y) - \cos(2\pi a))\bigr).$$

Consequently, with $z_t = \exp(2\pi i a - 2\pi b t)$,

$$S_a(b) + S_{1-a}(b) = 2\pi b\log 2 + 2\operatorname{Re}\int_0^1 \frac{\operatorname{Li}_1(z_t) - \operatorname{Li}_1(z_1)}{1-t^2}\,\mathrm{d}t. \tag{5.9}$$

To verify the constant, write the logarithm as $2\pi bt - 2\operatorname{Re}\operatorname{Li}_1(z_t)$ and use $\int_0^1 (1-t)/(1-t^2)\,\mathrm{d}t = \log 2$. The integral is ordinary and has a removable endpoint at one. At $a = 1/2$, this yields

$$S_{1/2}(b) = \pi b\log 2 + \int_0^1 \frac{\operatorname{Li}_1(-e^{-2\pi bt}) - \operatorname{Li}_1(-e^{-2\pi b})}{1-t^2}\,\mathrm{d}t. \tag{5.10}$$

Taking $a \downarrow 0$ in (5.9) gives

$$S_1(b) = -\frac{\pi^2}{8} + \pi b\log 2 + \int_0^1 \frac{\operatorname{Li}_1(e^{-2\pi bt}) - \operatorname{Li}_1(e^{-2\pi b})}{1-t^2}\,\mathrm{d}t. \tag{5.11}$$

Here the logarithmic singularity at $t = 0$ is integrable. On the series side $S_a(b) \to \pi^2/4 + S_1(b)$, which accounts for the otherwise easily missed term $-\pi^2/8$.

6 Second jets at every nonpositive integer: finite polylogarithmic integrands

Let $0 < a < 1$, $b > 0$ and

$$E(s;a,b) = \sum_{n\in\mathbb{Z}} \bigl((n+a)^2 + b^2\bigr)^{-s} = Z_1(s;a,b) + Z_1(s;1-a,b). \tag{6.1}$$

Put $L_m(a,b) = E'(-m;a,b)$ and $M_m(a,b) = E''(-m;a,b)$. The first-jet formula of the predecessor [2] is

$$\begin{aligned} L_m(a,b) = {} & -\frac{2^{2m+1}(m!)^2\pi}{(2m+1)!}b^{2m+1} \\ & + 2(-1)^m m!\sum_{j=0}^{m}\frac{(m+j)!\,b^{m-j}}{j!(m-j)!4^j\pi^{m+j}}\operatorname{Re}\operatorname{Li}_{m+j+1}(e^{-2\pi b+2\pi i a}). \end{aligned} \tag{6.2}$$

In particular,

$$L_0(a,b) = -\log\bigl(2(\cosh(2\pi b) - \cos(2\pi a))\bigr), \tag{6.3}$$

$$L_1(a,b) = -\frac{4\pi b^3}{3} - \frac{2b}{\pi}\operatorname{Re}\operatorname{Li}_2(z) - \frac{1}{\pi^2}\operatorname{Re}\operatorname{Li}_3(z). \tag{6.4}$$

These inherited formulas can be obtained from Poisson summation and the finite half-integer K-Bessel polynomial. They also satisfy $\partial_{b^2} L_m = mL_{m-1}$, with the zero-mass values fixed by the Hurwitz functional equation. The new second-jet theorem below uses their *finite explicit right sides*, not an unevaluated first-jet symbol.

Theorem 6.1 (All negative-point second jets)**.** *For every $m \geq 0$, $0 < a < 1$ and $b > 0$, define*

$$R_m(t; a, b) = \sum_{\ell=0}^{m} \binom{m}{\ell} (b^2(1-t^2))^{m-\ell} L_\ell(a, bt). \tag{6.5}$$

Then

$$\begin{aligned} M_m(a,b) = 4\,\mathrm{Re} \sum_{\alpha \in \{a, 1-a\}} \sum_{j=0}^{m} \binom{m}{j} (-2ib)^{m-j} \zeta''(-m-j, \alpha+ib) \\ - 4\int_0^1 \frac{L_m(a,b) - R_m(t;a,b)}{1-t^2}\,\mathrm{d}t. \end{aligned} \tag{6.6}$$

The integrand is explicit in finitely many ordinary polylogarithms via (6.2)*; the endpoint at $t = 1$ is removable and the endpoint at $t = 0$ is regular. No Bessel-order derivative or unspecified finite part occurs.*

Proof. Apply Theorem 4.1 at each of the two shifts with the fixed even polynomial $W_b(x) = (x^2 + b^2)^m$. Holding this polynomial fixed while differentiating the spectral variable gives $Z''_{W_b}(0; \alpha, b) = Z''_1(-m; \alpha, b)$ and $C_{W_b} = 0$. If $y = x + ib$, then

$$(x^2 + b^2)^m = y^m (y - 2ib)^m = \sum_{j=0}^{m} \binom{m}{j} (-2ib)^{m-j} y^{m+j}.$$

Therefore the linear endpoint contribution is exactly the finite sum in (6.6).

For an arbitrary fixed even polynomial W, the first defect vanishes, so

$$T_W(a, y) := Z'_W(0; a, y) + Z'_W(0; 1-a, y) = 2\,\mathrm{Re}\{\Phi_{a,W}(iy) + \Phi_{1-a,W}(iy)\}.$$

Equation (4.6) consequently gives

$$\mathcal{K}_{a,W}(ib, -ib) + \mathcal{K}_{1-a,W}(ib, -ib) = 4\int_0^1 \frac{T_W(a,b) - T_W(a,bt)}{1-t^2}\,\mathrm{d}t. \tag{6.7}$$

For $W = W_b$, expand instead around the intermediate mass bt:

$$(x^2 + b^2)^m = \sum_{\ell=0}^{m} \binom{m}{\ell} (b^2(1-t^2))^{m-\ell} (x^2 + b^2t^2)^\ell.$$

This proves $T_{W_b}(a, bt) = R_m(t; a, b)$ and $T_{W_b}(a, b) = L_m(a, b)$, hence (6.6).

At $t = 1$, $R_m(1) = L_m$ and all functions are differentiable, so the quotient has a finite limit. At zero the bilateral lattice has a positive gap because $0 < a < 1$; its continued jets are analytic in b^2 near zero. Equivalently, the polylogarithms at $e^{2\pi i a} \neq 1$ in (6.2) have regular limits. This proves the endpoint claims. □

For $m = 0$ this is the transparent formula

$$\begin{aligned} E''(0; a, b) &= 4\,\mathrm{Re}\{\zeta''(0, a+ib) + \zeta''(0, 1-a+ib)\} + 4S_a(b) + 4S_{1-a}(b) \\ &= 4\,\mathrm{Re}\{\zeta''(0, a+ib) + \zeta''(0, 1-a+ib)\} + 8\pi b \log 2 \\ &\quad + 8\,\mathrm{Re}\int_0^1 \frac{\mathrm{Li}_1(e^{2\pi i a - 2\pi bt}) - \mathrm{Li}_1(e^{2\pi i a - 2\pi b})}{1-t^2}\,\mathrm{d}t. \end{aligned} \tag{6.8}$$

This is a representation theorem, not a nonreducibility theorem. In particular, retaining an ordinary polylogarithmic integral is not evidence that no further finite reduction exists.

6.1 A normalized all-jet mass-antiderivative ladder

The second jets themselves satisfy an inhomogeneous relation:

$$\partial_\mu M_m(a,\sqrt{\mu}) = mM_{m-1}(a,\sqrt{\mu}) - 2L_{m-1}(a,\sqrt{\mu}), \qquad m \geq 1. \tag{6.9}$$

Harmonic normalization removes the inhomogeneous term. In fact the same normalization works at every spectral order.

Let $h_{m,q}$ be the complete homogeneous polynomial in $1, 1/2, \ldots, 1/m$, defined by

$$\sum_{q\geq 0} h_{m,q}t^q = \prod_{j=1}^{m}(1-t/j)^{-1} = \exp\left(\sum_{p\geq 1}\frac{H_m^{(p)}}{p}t^p\right). \tag{6.10}$$

For $m = 0$ the product is one. Define, for $r \geq 1$,

$$\mathcal{A}_{m,r}(a,\mu) = \partial_t^r\left\{\frac{m!}{(1-t)_m}E(t-m;a,\sqrt{\mu})\right\}\Big|_{t=0}. \tag{6.11}$$

Theorem 6.2 (Every normalized jet is an Appell-type mass ladder)**.** *For $0 < a < 1$, $\mu \geq 0$, $m \geq 1$ and $r \geq 1$,*

$$\mathcal{A}_{m,r}(a,\mu) = r!\sum_{j=1}^{r}\frac{h_{m,r-j}}{j!}E^{(j)}(-m;a,\sqrt{\mu}), \tag{6.12}$$

$$\partial_\mu \mathcal{A}_{m,r}(a,\mu) = m\mathcal{A}_{m-1,r}(a,\mu). \tag{6.13}$$

Consequently every anchored mass primitive is explicit:

$$\begin{aligned}\mathcal{A}_{m,r}(a,\mu) = &\sum_{j=0}^{m-1}\binom{m}{j}\mu^j\mathcal{A}_{m-j,r}(a,0)\\ &+ m\int_0^\mu(\mu-u)^{m-1}E^{(r)}(0;a,\sqrt{u})\,\mathrm{d}u.\end{aligned} \tag{6.14}$$

The boundary constants are finite Hurwitz jets because

$$E^{(j)}(-m;a,0) = 2^j\{\zeta^{(j)}(-2m,a) + \zeta^{(j)}(-2m,1-a)\}. \tag{6.15}$$

In particular, $\mathcal{A}_{m,1} = L_m$ and $\mathcal{A}_{m,2} = M_m + 2H_mL_m$.

Proof. The parameter identity $\partial_\mu E(s) = -sE(s+1)$ holds first in absolute convergence and then by continuation. Therefore, as germs in t,

$$\partial_\mu\left\{\frac{m!}{(1-t)_m}E(t-m)\right\} = m\frac{(m-1)!}{(1-t)_{m-1}}E(t-(m-1)).$$

Differentiate in t to obtain (6.13). The identities $E(-m) = 0$ follow from the binomial continuation and $\zeta(-2h,a) + \zeta(-2h,1-a) = 0$, the latter being Bernoulli reflection. Expanding the prefactor by (6.10) now gives (6.12). The positive lattice gap permits continuation and parameter differentiation through $\mu = 0$; at zero the identity $E(s;a,0) = \zeta(2s,a) + \zeta(2s,1-a)$ proves (6.15). Repeated integration of the ladder, with these boundary data, gives (6.14). □

Combining (6.14) at $r = 2$ with Theorem 6.1 evaluates every polynomially weighted mass integral of $E''(0)$ by explicit endpoints and a single finite polylogarithmic-integrand coordinate. No integration constants are chosen after the fact.

7 Several quadratic factors and the complete confluent expansion

Throughout this section $W(x) = P(x^2)$ is even; its coefficients may be complex. For $b_1, \dots, b_r > 0$ define

$$Z_{W;\mathbf{b}}(s;a) = \sum_{n\geq 0} W(x_n) \prod_{j=1}^{r} (x_n^2 + b_j^2)^{-s}. \tag{7.1}$$

The same notation with a superscript bil denotes the sum over the full lattice, equivalently the sum of the one-sided functions at a and $1-a$, when $0 < a < 1$.

For $\mu > 0$ put

$$\mathcal{T}_{a,W}(\mu) = Z'_W(0;a,\sqrt{\mu}) = \Phi_{a,W}(i\sqrt{\mu}) + \Phi_{a,W}(-i\sqrt{\mu}). \tag{7.2}$$

The last equality is the inherited first-defect parity formula. On the bilateral lattice it is the following finite polylogarithmic expression:

$$\mathcal{T}^{\mathrm{bil}}_{a,W}(\mu) = \sum_{\ell=0}^{\deg P} \frac{P^{(\ell)}(-\mu)}{\ell!} L_\ell(a,\sqrt{\mu}). \tag{7.3}$$

Thus the next definition uses already evaluated first jets only.

Definition 7.1 (Mass covariance)**.** For $\mu = b^2$, $\nu = c^2$, define

$$V_{a,W}(b,c) = \int_0^1 \frac{\mathcal{T}_{a,W}(\mu) + \mathcal{T}_{a,W}(\nu) - \mathcal{T}_{a,W}((1-t)\mu + t\nu) - \mathcal{T}_{a,W}(t\mu + (1-t)\nu)}{t(1-t)}\,\mathrm{d}t. \tag{7.4}$$

Use $\mathcal{T}^{\mathrm{bil}}$ for V^{bil}. The name denotes this explicitly defined analytic function; it does not imply a probability interpretation for arbitrary polynomial weights.

The integrand has removable endpoints and is identically zero when $b = c$. The formula is analytic in μ, ν on a neighborhood of every positive real rectangle, including the diagonal.

Theorem 7.2 (All second jets are pairwise mass covariances)**.** *For every even polynomial W, every $a > 0$, and all positive masses,*

$$\boxed{Z''_{W;\mathbf{b}}(0;a) = r\sum_{j=1}^{r} Z''_W(0;a,b_j) - \sum_{1\leq i<j\leq r} V_{a,W}(b_i,b_j).} \tag{7.5}$$

In particular,

$$Z''_{W;b,c}(0;a) = 2Z''_W(0;a,b) + 2Z''_W(0;a,c) - V_{a,W}(b,c). \tag{7.6}$$

Every term is explicit by Theorem 4.1 and (7.4)*. The bilateral statements follow by reflection and have finite polylogarithmic covariance integrands via* (7.3)*. Repeated masses are allowed; in particular $Z''_{W;b,b}(0;a) = 4Z''_W(0;a,b)$.*

Proof. We first prove the two-factor statement directly from a beta integral, without a formal manipulation of divergent logarithmic sums. Euler's identity gives

$$Z_{W;b,c}(s;a) = \frac{\Gamma(2s)}{\Gamma(s)^2} \int_0^1 t^{s-1}(1-t)^{s-1} Z_W\big(2s;a,\sqrt{(1-t)\mu + t\nu}\big)\,\mathrm{d}t$$

in its initial convergence half-plane. Subtract its endpoint average as in Theorem 2.2. For even W,

$$Z_W(0;a,b)=\sum_{d\ \text{even}} w_d\zeta(-d,a) \tag{7.7}$$

is independent of b: in the inverse-power expansion every nonconstant Pochhammer coefficient vanishes at zero, and $2k-d$ cannot equal one. Thus the subtracted zero jet is identically zero. The second beta row contains only the first-jet chord, exactly (7.4). This proves (7.6), including continuation and endpoint regularity.

For completeness, a common coefficient argument justifies the extension to all r. Choose a tail $x\ge X>\max b_j$ and expand each product as

$$x^{-2rs}\sum_{k\ge 0} c_k(s)x^{-2k}.$$

After multiplication by an even polynomial W, continuation replaces each power by $\zeta(2rs+2k-d,X)$. None of these Hurwitz terms has a pole at $s=0$, since $2k-d$ is even. Therefore taking two spectral derivatives is equivalent to applying the same linear coefficient rule

$$x^e(\log x)^q \longmapsto (-1)^q\zeta^{(q)}(-e,X),\qquad e\ \text{even},$$

to the pointwise second-derivative expansion. This rule is independent of the initial scale $2r$; that scale is already accounted for by the powers of $\log x$. The expansions converge normally after the finite head, as in Theorem 3.1.

Apply this common linear rule to the elementary identity

$$\left(\sum_{j=1}^{r} A_j\right)^2 = r\sum_{j=1}^{r} A_j^2-\sum_{i<j}(A_i-A_j)^2,\qquad A_j=\log(x^2+b_j^2).$$

The two-factor case has already identified the image of $(A_i-A_j)^2$ as $V_{a,W}(b_i,b_j)$. The finite head satisfies the same pointwise identity. This proves (7.5) without assuming linearity across a hidden pole. Reflection proves the bilateral case. The coincident statement follows either from $V(b,b)=0$ or from $Z_{W;b,b}(s)=Z_W(2s;b)$. □

7.1 An independent Gamma polarization of the covariance

There is a second useful expression for the same function:

$$\begin{aligned}V_{a,W}(b,c)=&\sum_{\epsilon,\eta\in\{1,-1\}}\mathcal{K}_{a,W}(\epsilon ib,\eta ic)\\&-\mathcal{K}_{a,W}(ib,-ib)-\mathcal{K}_{a,W}(ic,-ic).\end{aligned} \tag{7.8}$$

To prove it, first use a tail with both masses small relative to x. The algebraic combination of logarithm squares on the right is $\log^2((x^2+b^2)/(x^2+c^2))$. In the expansion (4.7), the extra harmonic term at degree k is proportional to w_{k-1}. The combined logarithm-square expression has only even inverse powers, whereas even W can contribute to w_{k-1} only for odd k. Hence all such harmonic corrections cancel. The common coefficient argument in the preceding proof identifies the remaining expression with V. Analytic continuation in positive masses proves the global identity. Formula (7.8) is not needed to define V and is valuable as an independent evaluation route.

When $\deg W \le 2$, the covariance is the ordinary convergent series

$$V_{a,W}(b,c) = \sum_{n\ge 0} W(x_n) \log^2\left(\frac{x_n^2+b^2}{x_n^2+c^2}\right), \qquad W \text{ even}, \quad \deg W \le 2. \tag{7.9}$$

Indeed, the logarithm square is $O(x^{-4})$. For higher polynomial degree the unmodified series generally diverges, and (7.4), not that divergent series, is the assertion.

For the unweighted bilateral lattice, define $D(a,y) = 2(\cosh(2\pi y) - \cos(2\pi a))$. The following ordinary identity is an explicit elementary-integrand specialization:

$$\begin{aligned}&\sum_{n\in\mathbb{Z}} \log^2\left(\frac{(n+a)^2+b^2}{(n+a)^2+c^2}\right)\\ &\quad = \int_0^1 \frac{1}{t(1-t)} \log\left(\frac{D(a,\sqrt{(1-t)b^2+tc^2})D(a,\sqrt{tb^2+(1-t)c^2})}{D(a,b)D(a,c)}\right) \mathrm{d}t.\end{aligned} \tag{7.10}$$

All quantities are positive inside the logarithms, so no branch convention is implicit.

7.2 Every confluent coefficient is a finite polygamma expression

Theorem 7.3 (Complete coincident-mass expansion)**.** *Let $a, b > 0$ and W be even. For h sufficiently small, more precisely $|h| < a^2 + b^2$ in the one-sided analytic squared-mass continuation,*

$$\boxed{V_{a,W}(b, \sqrt{b^2+h}) = \sum_{k\ge 2} \frac{2(-1)^k H_{k-1}}{k}\, h^k Z_W(k;a,b).} \tag{7.11}$$

The left side denotes analytic continuation in h of its positive-mass germ. The square-root notation creates no singularity at $h = -b^2$, because the spectrum is even in the mass. The series is locally normally convergent. On the bilateral lattice, replace the radius by $\min(a, 1-a)^2 + b^2$ and use the bilateral zeta. All displayed positive-integer values are holomorphic continued values, and each has a finite polygamma–Bernoulli evaluation. The first terms are

$$h^2 Z_W(2;a,b) - h^3 Z_W(3;a,b) + \frac{11}{12} h^4 Z_W(4;a,b) + O(h^5). \tag{7.12}$$

Proof. The even-weight continuation has no positive-integer spectral poles: its possible poles are at half-integers, as is immediate from the inverse-power expansion. Repeated mass differentiation of the first jet therefore gives

$$\mathcal{T}^{(k)}_{a,W}(\mu) = (-1)^k (k-1)! Z_W(k;a,\sqrt{\mu}), \qquad k \ge 1. \tag{7.13}$$

Expand the ordinary analytic function $\mathcal{T}$ in (7.4) about $\mu = b^2$. The coefficient of h^k is

$$\frac{\mathcal{T}^{(k)}(\mu)}{k!} \int_0^1 \frac{1 - t^k - (1-t)^k}{t(1-t)}\, \mathrm{d}t = \frac{2H_{k-1}}{k!} \mathcal{T}^{(k)}(\mu).$$

The elementary integral follows by finite geometric summation. This proves the coefficients. The nearest zero of $x_n^2 + \mu + h$ is at $h = -a^2 - b^2$; the finite-head continuation is analytic in the open disk bounded by that point. The Taylor expansion and the endpoint subtraction are locally uniform on smaller disks, proving normal convergence. The bilateral radius follows from the smallest lattice gap. □

Here is the promised finite evaluation of the coefficients. For $k \geq 1$,

$$Z_1(1; a, b) = \frac{\psi(a + ib) - \psi(a - ib)}{2ib}, \tag{7.14}$$

$$Z_1(k; a, b) = \frac{(-1)^{k-1}}{(k-1)!} \left(\frac{\partial}{\partial \mu}\right)^{k-1} \frac{\psi(a + i\sqrt{\mu}) - \psi(a - i\sqrt{\mu})}{2i\sqrt{\mu}}\bigg|_{\mu=b^2}. \tag{7.15}$$

Only finitely many ordinary polygamma derivatives occur. For $m \geq 0$,

$$Z_1(-m; a, b) = \sum_{j=0}^{m} \binom{m}{j} b^{2(m-j)} \zeta(-2j, a), \tag{7.16}$$

whose Hurwitz values are Bernoulli polynomials. Expanding the weight around $-b^2$ gives, for every positive integer k,

$$Z_W(k; a, b) = \sum_{\ell=0}^{\deg P} \frac{P^{(\ell)}(-b^2)}{\ell!} Z_1(k - \ell; a, b). \tag{7.17}$$

Equations (7.15) and (7.16) cover all terms. No confluent denominator needs to be evaluated by subtracting nearby masses.

A related mixed derivative identity is

$$\partial_\mu \partial_\nu Z''_{W;\sqrt{\mu},\sqrt{\nu}}(0; a) = 2\frac{Z_W(1; a, \sqrt{\mu}) - Z_W(1; a, \sqrt{\nu})}{\nu - \mu}, \tag{7.18}$$

with diagonal value $2Z_W(2; a, \sqrt{\mu})$. To prove it, differentiate the initial spectral series twice in the masses, obtaining a factor s^2, and then take the second spectral derivative. Rational partial fractions give the right side. The same even-power pole exclusion used above justifies this continuation. Thus both the covariance and its mass Hessian have completely specified, regular diagonal limits.

8 Anchored shift primitives of every spectral order

The preceding representations also admit a uniform antiderivative calculus in the Hurwitz shift a. This is separate from the mass ladder: the two parameters should not be interchanged informally.

For integers $R \geq 1$, $h, j \geq 0$, define

$$\mathcal{P}^{[j]}_{R,h}(a) = (-1)^R \, \partial_s^j \left(\frac{\zeta(s - R, a)}{(s - R)_R}\right)\bigg|_{s=-h}. \tag{8.1}$$

The denominator has no zero at $s = -h$; all its factors are strictly negative integers there. Thus the formula is a finite combination of negative-integer Hurwitz derivatives and generalized harmonic numbers, with no resonance prescription needed. Its basic property is

$$\partial_a^R \mathcal{P}^{[j]}_{R,h}(a) = \zeta^{(j)}(-h, a). \tag{8.2}$$

For the Gamma case $j = 1$, this reads explicitly

$$\mathcal{P}^{[1]}_{R,h}(a) = \frac{h!}{(h + R)!} \left\{\zeta'(-h - R, a) + (H_{h+R} - H_h)\zeta(-h - R, a)\right\}. \tag{8.3}$$

For example, $\partial_a\{\zeta'(-1,a)+\zeta(-1,a)\}=\zeta'(0,a)$. The elementary $\frac{1}{2}\log(2\pi)$ can be added when a primitive of $\log\Gamma(a)$ rather than $\zeta'(0,a)$ is desired.

To fix all constants, let $a_0>0$ and write

$$\mathfrak{A}_{a_0,R}Q(a)=Q(a)-\sum_{q=0}^{R-1}\frac{(a-a_0)^q}{q!}Q^{(q)}(a_0). \tag{8.4}$$

This is the unique Rth primitive of $Q^{(R)}$ whose derivatives of orders $0,\dots,R-1$ vanish at a_0.

Theorem 8.1 (All anchored shift primitives of the defect)**.** *For fixed admissible endpoints u,v and polynomial W, every $\Delta_r(a,W;u,v)$ has an explicitly anchored Rth primitive on any positive a-interval where the endpoint segments remain admissible. In Theorem 2.3, replace each finite endpoint function*

$$F^{(j)}_{a,W}(0;z)=\sum_{d=0}^{D}w_d\sum_{h=0}^{d}\binom{d}{h}(-z)^{d-h}\zeta^{(j)}(-h,a+z) \tag{8.5}$$

by

$$\mathfrak{A}_{a_0,R}\left[\sum_{d=0}^{D}w_d\sum_{h=0}^{d}\binom{d}{h}(-z)^{d-h}\mathcal{P}^{[j]}_{R,h}(a+z)\right]. \tag{8.6}$$

Keep the beta coefficients and logarithmic weights unchanged. The result is the required anchored primitive, still a finite sum of ordinary convergent integrals.

Proof. The classical parameter identity $\partial_a^R\zeta(s-R,a)=(-1)^R(s-R)_R\zeta(s,a)$ proves (8.2) by differentiation in s. Differentiating the reciprocal finite product at $s=-h$ gives (8.3). The operator (8.4) imposes the stated initial conditions without changing the Rth derivative. The endpoint subtraction commutes with this operation because it is linear and the endpoints are fixed. On compact admissible a-intervals the endpoint estimates in Theorem 2.2 are uniform, including the fixed logarithmic weights. Therefore the R shift derivatives may pass under every integral. The resulting Rth derivative is precisely (2.18), and all lower derivatives at the anchor vanish. □

In the conjugate second-order case there is a shorter equivalent formula. Construct $\Phi^{\langle R\rangle}_{a,W}(z)$ by replacing the first jets in (2.4) with the anchored version of (8.3). Then

$$\int_{a_0}^{a}\frac{(a-x)^{R-1}}{(R-1)!}\Delta_2(W;x,b)\,\mathrm{d}x=\int_0^1\frac{\mathcal{C}_t\Phi^{\langle R\rangle}_{a,W}(ib,-ib)}{t(1-t)}\,\mathrm{d}t-C_W(b)\frac{(a-a_0)^R}{R!}. \tag{8.7}$$

The formula also holds for $a<a_0$ with oriented integration. It explicitly retains the primitive of the local polynomial, which must not be dropped merely because that polynomial is independent of a.

8.1 Parameter differentiation and generalized Stieltjes functions

The coefficient formula (3.11) is locally normal in a,b after a sufficiently large fixed head is selected on a compact parameter set. It can therefore be differentiated in either parameter any prescribed finite number of times. At resonant coordinates use (3.1) as a whole germ, rather than differentiating a singular isolated zeta term. In particular,

$$\partial_a\zeta(s,a)=-s\zeta(s+1,a)$$

converts the resulting derivatives into polygamma and Stieltjes/Hurwitz jets with explicit finite product coefficients. At an integer shift, $\zeta(k, a)$ is the corresponding generalized harmonic tail for $k \geq 2$. Thus the finite beta representation and the harmonic/Stieltjes series remain equal after all these legitimate differentiations and after the anchored integrations just proved.

9 Verification, correction guards, and integration status

9.1 What was actually checked

The mathematical theorems above are established by their analytic proofs. The accompanying computations check finite algebra and implementations; none is substituted for an all-parameter proof.

The exact suite passed **432 assertions** over rational polynomial algebra and formal Stieltjes coefficients. Its groups are:

Finite algebra checked	Assertions
Double-zero and duplication polynomials	60
Harmonic coefficients and Gamma corrections	60
Local beta moments through the required logarithmic order	32
Confluent log-square coefficients and beta moments	62
Shift-primitive normalization	128
All-jet mass-germ normalization	48
Explicit Stieltjes pole-jet extraction	42
Total	432

The tests include D_k through $k = 20$, the two local beta moments through $k = 16$, the confluent coefficient identity through $k = 32$, anchored primitive coefficients for $1 \leq R \leq 8$ and $0 \leq h \leq 7$, and the mass germ identity through $m = 24$. The complete finite assertions are recorded in `results/exact_results.json`.

The numerical suite completed **87 comparisons at 35 decimal-digit working precision**. The acceptance threshold was 10^{-23} for $|\text{lhs} - \text{rhs}|/\max(1, |\text{lhs}|, |\text{rhs}|)$. The largest recorded scaled residual was approximately 9.18×10^{-35}. This is a residual, not an error bound. The checks cover all of the following mechanisms: independent direct Pochhammer and double-zero defect jets through spectral order four; odd and even Gamma corrections; cutoff changes and masses outside the unshifted Taylor disk; ordinary logarithm and arctangent sums; reflected polylogarithmic integrals including a logarithmic endpoint; second jets at four nonpositive spectral points and two shift/mass pairs; two distinct Gamma/mass covariance representations; independently expanded two-factor spectra; repeated masses; finite polygamma confluence; shift primitives; and logarithmically weighted beta jets.

Some comparisons share the same underlying numerical Hurwitz-zeta implementation. They must not be advertised as entirely independent special-function libraries. The ordinary sums, Gamma products, polylogarithmic integrands and spectral expansions nevertheless provide different algebraic and analytic evaluation routes. No interval enclosures, certified infinite tails, or proof-assistant verification are included. Execution was completed in staged invocations; the final JSON contains every accepted comparison, not an extrapolated count.

9.2 Corrections versus safeguards

No false theorem was identified in the inspected predecessor's first-derivative statements. In particular it explicitly cautions that higher derivatives do not automatically factorize. The following are therefore proposed *integration safeguards*, not accusations that the predecessor made those errors.

Do not extend first-order parity to the full second jet. Even weights remove C_W and the $I_{0,p}$ terms, but not $\mathcal{K}$. The exact counterexample is $\Delta_2(1; a, b) = -4S_a(b) < 0$ for $b > 0$. Theorem 4.2 gives the full universal criterion.

Retain one extra coefficient before a pole multiplication. The resonant rth derivative needs $d_{k,r+1}$, not only $d_{k,r}$. Equation (3.10) is the exact replacement rule. This is also why generalized Stieltjes coefficients and ordinary zeta derivatives must not be mixed without tracking the spectral scale two.

Do not silently remove the Gamma correction for odd weights. For $W = x$ the second defect is $\Delta_2(x; a, b) = \mathcal{K}_{a,x}(ib, -ib) + 4b^2$. Omitting the $4b^2$ term is an explicitly false identity.

Normalize the second mass ladder. The pure ladder belongs to $M_m + 2H_m L_m$, not to M_m alone. At all orders the correct object is the germ $m!E(t - m)/(1 - t)_m$. The primitive constants are those in (6.15), not arbitrary polynomials.

Distinguish continuation from an ordinary sum. The covariance series in (7.9) is ordinary for even weights of degree at most two. At higher degree use the ordinary *integral* definition (7.4); the original logarithmic series is generally divergent. Similarly, a small-mass series is not globally valid without its finite head.

Preserve the branch and endpoint subtractions. Use analytic $\log\Gamma$, retain the paired numerator under the integral, and evaluate the diagonal by its confluent formula. Integrating its four separated divergent endpoint terms is not equivalent to the subtracted beta integral.

9.3 Suggested repository integration

A suitable new report destination is

> `Analysis/Polylogarithms/docs/reports/stieltjes-correlation-calculus/continuations/higher-spectral-defects/`.

The package includes a compact theorem fragment with the prefix `hsd:`, source hashes, a machine-readable claim ledger, and an editorial status note. The predecessor's Section 7.6 Questions 2 and 7 can be cross-referenced to the explicit representation theorems here. Question 1 can record the new finite-integrand representation while retaining its minimum-function-algebra component as open. The classical first-derivative results should keep their original attribution. No Gaussian special-value status or numerical independence status should be changed. No remote commit, overwrite, or repository modification was performed in producing this package.

10 Further research questions and concrete next targets

1. Finite closure of the negative-point second integrals. Determine special parameter families for which the integral in (6.6) reduces to a finite collection of polylogarithms, polylogarithm order derivatives, and explicit logarithms. A useful first target is the half-shift $a = 1/2$ at $m = 0, 1$. A failure in one prescribed symbolic calculus is not a proof of analytic nonreducibility.

2. A finite polylogarithmic dictionary for every higher jet. Theorem 2.3 lowers the inner Hurwitz order by one. Can reflection and Fourier continuation turn those lower jets into a uniform finite set of logarithmically weighted polylogarithm-order integrals at every r? The goal should include the exact endpoint constants and not merely a formal interchange of a Fourier series with a singular beta weight.

3. Higher-order multi-mass connected terms. The pairwise formula (7.5) relies on a quadratic variance identity. At third and higher order, identify the appropriate three-point and higher endpoint subtractions in a Dirichlet-simplex parameterization. Determine which terms genuinely require more than pairwise covariances and how their coincident limits encode products of harmonic coefficients and higher resolvents.

4. Multi-mass formulas for arbitrary odd polynomial weights. For odd weights the zero quadratic jet depends polynomially on the mass, so the local term in the mass beta formula need not vanish. Compute its all-order residue polynomial and the corresponding corrected multi-mass covariance formula. The all-order linear-factor beta master and the explicit J_k moments provide exact interpolation constraints.

5. Arithmetic and colored weights. Replace the polynomial weight by a polynomial times a periodic sequence. Separate its mean, which controls the arithmetic Hurwitz pole, from the nontrivial character components. Seek a small-denominator Gamma/Dirichlet L dictionary for the resulting beta polarizations and their Stieltjes jets, while keeping analytic evaluation separate from claims of period independence.

6. Joint lattice-gap and mass degeneration. The formulas are regular for $0 < a < 1$ and positive mass, but the corner $(a, b) = (0, 0)$ has a collapsing lattice point. Construct a two-parameter subtracted germ that separates that point explicitly. Determine which normalized mixed jets have path-independent limits and which retain specified logarithmic path terms.

7. Nonintegral polynomial degree. For weights x^ρ, the finite shifted-Hurwitz expression (2.3) is no longer available. Develop an analytic replacement that interpolates the odd-degree residue corrections as ρ crosses integers. A correct interpolation must reproduce both Δ_1 and C_W at all polynomial weights, rather than assuming the second correction is a continuation of first-order parity.

8. Formalization and validated evaluation. The beta subtraction, polynomial pole extraction, harmonic recurrences, and normalized primitive identities have clean finite algebraic layers. Formalize those first, then the normal-convergence and differentiation lemmas. Independently, equip the

ordinary beta/Gamma integrals and finite-head tails with interval error control. Such a validated evaluator would strengthen the supplied diagnostics without changing the logical role of the analytic proofs.

Conclusion

The second spectral derivative is not governed by the first-derivative parity law. Its nonlocal part is nevertheless explicit: a Gamma–Hurwitz polarization, with a finite harmonic correction for odd weights. Endpoint-subtracted beta parameterization extends this mechanism to every spectral order and lowers the inner order by one. The independent harmonic–Stieltjes series supplies every resonant coefficient and a globally valid finite-head implementation.

The same construction controls several quadratic masses, gives regular confluent polygamma expansions, and converts the predecessor's finite first-jet polylogarithms into explicit second-jet integrands. Harmonic normalization then produces an all-jet mass-antiderivative ladder, while negative Hurwitz derivatives fix every shift-primitive constant. These are exact representation and structural theorems. The remaining questions concern further finite closure, arithmetic specialization, higher connected terms, and formal or interval certification, not unreported conjectural steps in the identities proved here.

References

[1] V. Reshetnikov and ProveIt contributors, *ProveIt: Polylogarithms manuscript*, canonical preface, differentiation chapter, research-status chapter, and incoming-report catalogue, inspected 11 October 2026. Repository checkpoint `5a5c0ed6573d25fcabf93653d72449f76df2df79`. `https://github.com/VladimirReshetnikov/ProveIt`. The source audit distinguishes preliminary main-branch reads from the byte-identified predecessor below.

[2] *Quadratic-Spectrum Stieltjes Identities: Complete reflected jets, harmonic counterterms, finite polylogarithmic reductions, and a polynomial parity criterion*, ProveIt research continuation, 10 October 2026. Especially Sections 5–7 and Section 7.6 Questions 1, 2, and 7. Byte-identified single-file source at `Analysis/Polylogarithms/docs/reports/stieltjes-correlation-calculus/continuations/quadratic-spectrum-stieltjes/article_singlefile.tex`, repository checkpoint `5a5c0ed6573d25fcabf93653d72449f76df2df79`. SHA-256 recorded in Section 1. The predecessor's own source anchor is `446dca9bd65c20e409d1d2561d4ffad408103e2a`.

[3] ProveIt, *Sixteenth incoming-archive manifest*, `Analysis/Polylogarithms/docs/manuscript/verification/sixteenth-incoming-archives.json`, checkpoint `5a5c0ed6573d25fcabf93653d72449f76df2df79`. Consulted for the predecessor's path, byte length, and SHA-256; not as a substitute for reading its mathematical text.

[4] NIST Digital Library of Mathematical Functions, §5.12, *Beta function*, especially Euler's integral and Eq. 5.12.4. `https://dlmf.nist.gov/5.12`. Consulted 11 October 2026. The two-denominator parameterization used here is a direct specialization.

[5] NIST Digital Library of Mathematical Functions, §5.8, *Infinite products*; §5.5, *Functional relations*; and §5.15, *Polygamma functions*. `https://dlmf.nist.gov/5.8`; `https://dlmf.nist.gov/5.5`; `https://dlmf.nist.gov/5.15`.

[6] NIST Digital Library of Mathematical Functions, §25.11, *Hurwitz zeta function*: continuation, parameter derivatives, negative-integer values, and the derivative at zero. `https://dlmf.nist.gov/25.11`. Consulted 11 October 2026.

[7] NIST Digital Library of Mathematical Functions, §25.12, *Polylogarithms*. `https://dlmf.nist.gov/25.12`. Consulted 11 October 2026.

[8] J. S. Dowker, *Calculation of the multiplicative anomaly*, arXiv:1412.0549, 2014. `https://arxiv.org/abs/1412.0549`. Cited for the classical first-derivative anomaly setting and factorization scope. The endpoint-subtracted higher-jet identities above are proved with their own specified normalization.